Introduction

The practicals of this assignment will be developed using the RelBot. The RelBot can be connected to a laptop via a dedicated internal network (see setup document). The communication can be performed using a ROS2 node that allows receiving the images and provides the commands to guide the robot during its missions. Communication through a customized Wi-Fi network guarantees limited delays and real-time capabilities to execute missions.

Please note: *Each group will be allowed to book the time slots to use the robots.*

Assessment of each assignment

The assignment should deliver a short video/presentation (called *video report* in the following) recorded for each assignment. The group should deliver this output (combination of images, videos, and slides) to explain what they implemented and how their solution performs in reality, adding footages of their (most successful) tests. The maximum length of each video (one video per assignment) is **3 minutes** and no more than **20 MB**. In total, the three assignments (described below) lead to a unique video (**max duration 9 minutes**) that must be delivered before the deadline.

The video should be technically informative: details on the algorithms used, workflows and other relevant elements must be clearly reported. In addition, the video report should evidence the capabilities of the developed solution, showcasing tests, also in “challenging” scenarios.

Assignment 1.1 – Object detection and tracking

The group will initially get familiar with the RelBot, learn how to communicate with it, receive images from the robot and send commands for its control through ROS2 (see setup document and GitHub page). In this assignment, each group is supposed to implement an object detection and tracking algorithm on the laptop and use it to detect a person walking in front of the RelBot and moving to its left and right. The group will also have to fine-tune the algorithm to detect the helmets on the people's heads and track them too. The second task of the assignment is to start a basic control of the RelBot, asking it to follow the person (without moving) and turn left and right, keeping him/her in the center of the image.

The additional task (optional, 1 extra point) can be to assess the robustness of the algorithm in the presence of multiple persons at the same time and partial occlusions (e.g. a person beyond a pillar) of the scene. For this extra task, the robot should be able to track persons with and without helmets: in other words, the robot should track a person, not just because he/she is the only one wearing the helmet.

This assignment should demonstrate real-time capabilities to be showcased in the video report.

Assignment 1.2 – Follow me

The group will have to implement the "follow-me" task on the RelBot. The robot should be able to follow one person, keeping a constant distance from him/her. A minimum length of **20 meters**, with at least **3 turns**, should be performed during the test. To do this, both object detection and tracking, as well as Single Image Depth Estimation (SIDE) algorithms, will need to be implemented and combined to retrieve the needed information.

An additional task (optional, 1 extra point) will be a basic collision avoidance algorithm where the robot should avoid small obstacles on the way, or even be able to turn a corner to follow the person who turned behind it.

This assignment should demonstrate real-time capabilities and be showcased in the video report.

Assignment 1.3 – SLAM

Every Group will read a scientific paper on SLAM from the given list below:

Assignment list and description: AI for Autonomous Robots (2025-2B)

Deliver a 2-minute video summarizing the workflow, the main achievements and comparing it with similar approaches in the literature.

An additional task (optional, 1 extra point) will be implementing a SLAM algorithm on the laptop and will be running to generate a 3D reconstruction of the environment using the images acquired by the RelBot during the "follow-me" assignment. Given the wide range of algorithmic options and the different processing requirements, no real-time capabilities must be demonstrated in this assignment. A justification of the typology of the algorithm and the processing time should be given in the video report. If you opt to take this challenging task, the total video can be extended to 3 minutes.

The group assignment 1 deadline is on 26/05/2024 for all the students.